

PAPER_720: Doc 43: Universal Permanence Equation, AGN Feedback, and the Final Parsec Problem

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Class: UQFFKnowledgeBaseKB5 **CP4 Entry:** #304 **Keywords:** UQFF, Universal Permanence, AGN feedback, Final Parsec, SMBH binary, non-local jump, metal retention, f_Z **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

Abstract

Doc 43 (UFE ORB EXP 2_24_07Mar2025) presents the Universal Permanence (UP) equation, a master UQFF sum including all field contributions (U_{gi} , U_m , TRZ zone, QS factor), applied to three astronomical phenomena: the Red Dwarf plasma orb (Drawing 30, Bear-den TRZ), AGN feedback mass-metallicity relation (Drawing 31, Sanchez et al.), and the Final Parsec problem for SMBH binaries. The non-local jump probability $P \approx 0.49/s$ demonstrates emergent quantum jumps in the UQFF framework.

1. Universal Permanence Equation

$$\text{UP}(t) = \sum_i k_i U_{gi} + \sum_j \frac{\mu_j}{r_j} (1 - e^{-\gamma t^-} \cos(\pi t_n)) \hat{\phi}_j U_{mj} + T_s^{\mu\nu} + U_b + NN + QS + ACE + DCE + [SSq] +$$
$$\approx 1.03 \times 10^{20} \cdot QS$$

where:

$$t^- = -t_n \cdot e^{\pi - t_n} \quad (t_n = 13.68 \text{ s} \implies t^- \approx -3.75 \times 10^{-4} \text{ s})$$

2. Non-Local Jump Probability

$$P = 1 - e^{-\gamma |t^-|} \approx 0.490 \text{ s}^{-1}$$

with $\gamma = 10^3 \text{ s}^{-1}$. Observed: 1-1.5 non-local jumps per video frame.

3. Energy Density

$$\rho_{react}(t) = 10^{15} \cdot e^{-0.001 \cdot t_n} \approx 9.86 \times 10^{14} \text{ W/m}^3$$

4. Metal Retention Factor f_Z

Galaxy Type	f_Z
Over-massive SMBH	0.89
Under-massive SMBH	0.85
Star-forming disk	0.73
Quenched disk	0.51

$$f_{CGM} = \frac{M_{CGM}}{M_{vir}}$$

5. Final Parsec

$$U_{g^4}^{FP}(t) = k_4 \cdot \rho_{SCm} \cdot \frac{M_{BH}^{binary}}{d_{parsec}} \cdot e^{-\alpha t} \cdot \cos(\pi t_n)$$

$M_{BH}^{binary} = 8.15 \times 10^{36} \text{ kg}, d_{parsec} = 2.55 \times 10^{20} \text{ m}:$

$U_{g^4}^{FP}(0) \approx 7.64 \times 10^{-6} \text{ J/m}^3$

6. AGN Feedback

$$U_{g^4}^{AGN}(t) \approx 8.40 \times 10^{-6} \cdot e^{-0.001t} \cdot \cos(\pi t_n) \text{ J/m}^3$$

for $\Delta M_{BH} = 1 \text{ dex}, f_{feedback} \approx 0.1.$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- UQFF Doc 43 (UFE_ORB_EXP_2_24_07Mar2025.docx)
- grok_share_ba508f76c8e.txt entry #69
- Sanchez et al. AGN feedback mass-metallicity relation
- Session 176, v5.33

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